https://github.com/am32-firmware/AM32/blob/main/Inc/targets.h

F051 Group B

TIE UNUSED NRESET HIGH

ESC x1 (NEED TO COPY X4)

Need temp sensor? or activate from imu?

Check if need more 47uF caps, compare to speedybee and other designs and ask why between BATT and GND – drawing uses 7 caps, speedybee uses 21

CAA572X7R1H476M670LJ

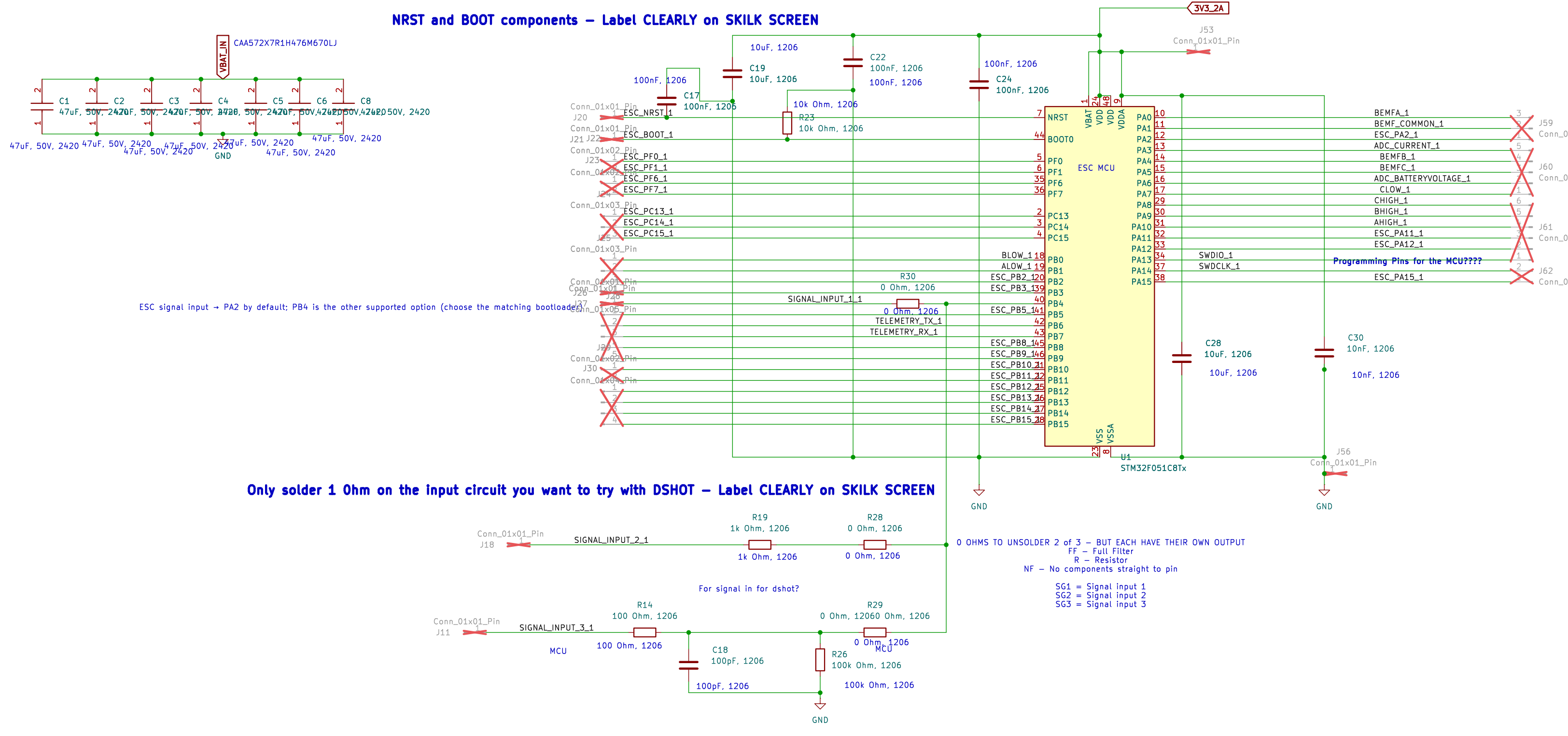
is this low ESR?

CL10A474KB8NNNC

The boot loader is located in System Memory. It is used to reprogram the Flash memory by
using USART on pins PA14/PA15 or PA9/PA10

LABEL PINS CLEARLY ON PCB

NRST and BOOT components – Label CLEARLY on SKILK SCREEN



ESC signal input → PA2 by default; PB4 is the other supported option (choose the matching bootloader)

Only solder 1 Ohm on the input circuit you want to try with DSHOT – Label CLEARLY on SKILK SCREEN

For signal in for dshot?

0 OHMS TO UNSOLDER 2 of 3 – BUT EACH HAVE THEIR OWN OUTPUT
FF – Full Filter
R – Resistor
NF – No components straight to pin

SG1 = Signal input 1
SG2 = Signal input 2
SG3 = Signal input 3

Mechanical Mounting Hole

CURRENT BUS BAR

https://www.digikey.co.uk/en/products/detail/altech-corporation/3803231/8547418

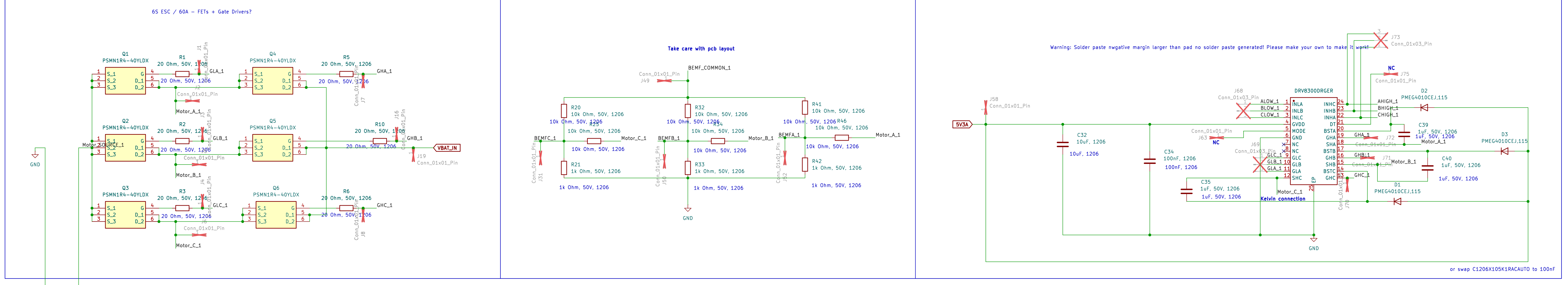
https://www.digikey.co.uk/en/products/detail/weidm-ller/0348900000/496908

https://www.amazon.co.uk/BENECREAT-Copper-Braided-Grounding-Reducing/dp/B0BGWZ5DGV?source=ps-sl-shoppingads-lpcontext&smid=A874XMFQVNIRI&th=1

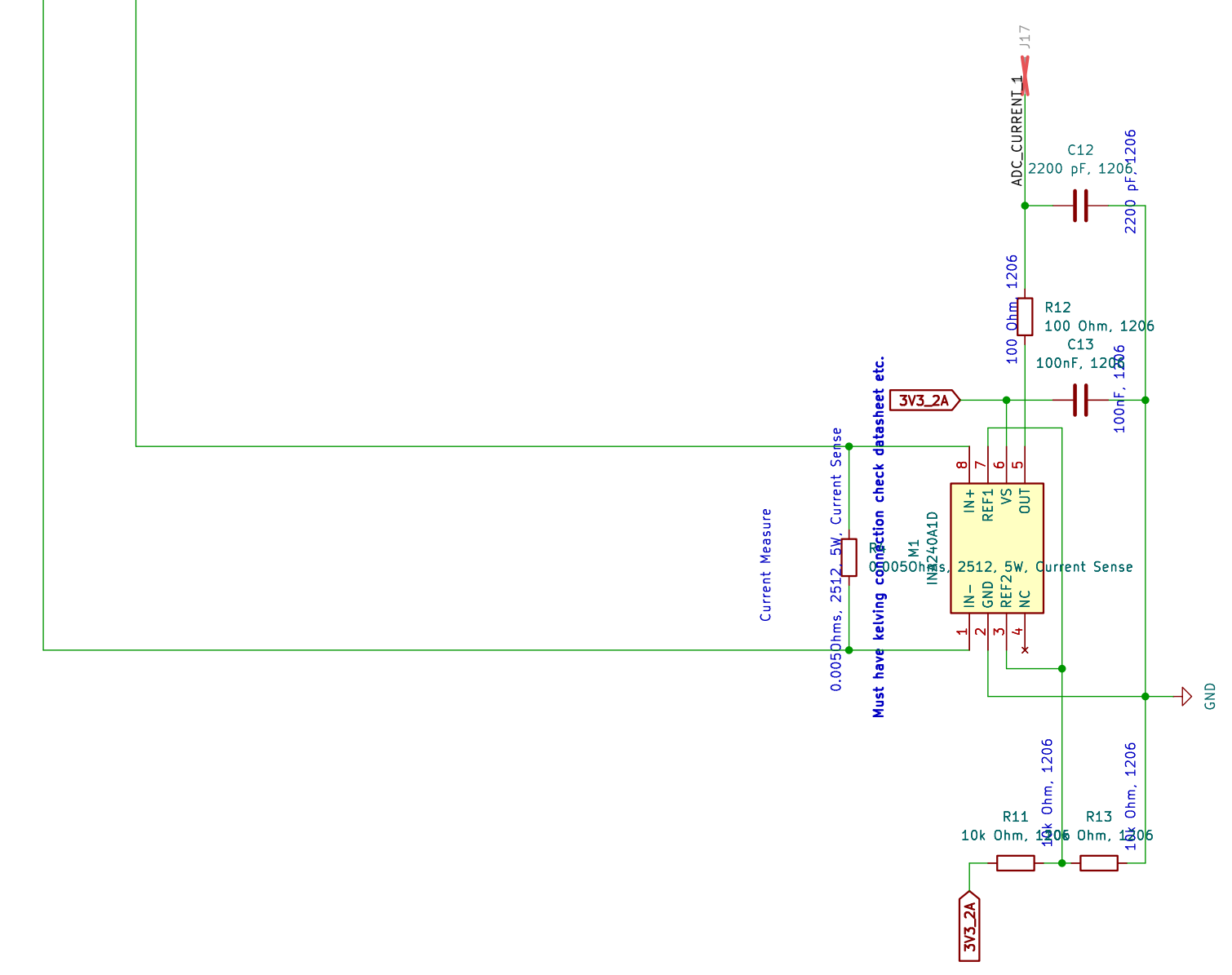
6S ESC / 60A – FETs + Gate Drivers?

Take care with pcb layout

Warning: Solder paste nwgative margin larger than pad no solder paste generated! Please make your own to make it work!



or swap C1206X105K1RACAUTO to 100nF



LABEL PASSIVE COMPONENTS

In an ESC design like AM32, those 7× 47 µF capacitors from VBAT to GND are the main bulk-decoupling network for the high-current DC bus.
Here's why so many–and why it's necessary:

1. Absorbing Current Spikes from the MOSFET Bridge

Each time a MOSFET phase leg switches, the motor windings create very fast di/dt pulses.

The battery leads have inductance; without local capacitance the voltage would ring or overshoot, potentially destroying the MOSFETs and MCU.

Multiple low-ESR electrolytics or polymer caps right at the FETs supply these instantaneous currents.

2. Reducing ESR and ESL

A single large capacitor has higher Equivalent Series Resistance (ESR) and Equivalent Series Inductance (ESL).

Paralleling several caps divides ESR/ESL:

ESR_total ≈ ESR_single / N

ESL_total ≈ ESL_single / N

Lower ESR means lower ripple voltage and less heating.

3. Handling Ripple Current & Heat

Each cap has a ripple-current rating.

Spreading the ripple across 7 parts keeps each one within spec and prolongs life.

4. Distributing Capacitance Physically

Placing several smaller capacitors along the DC bus keeps the impedance low across the whole board, not just at one point.

Typical Rule of Thumb

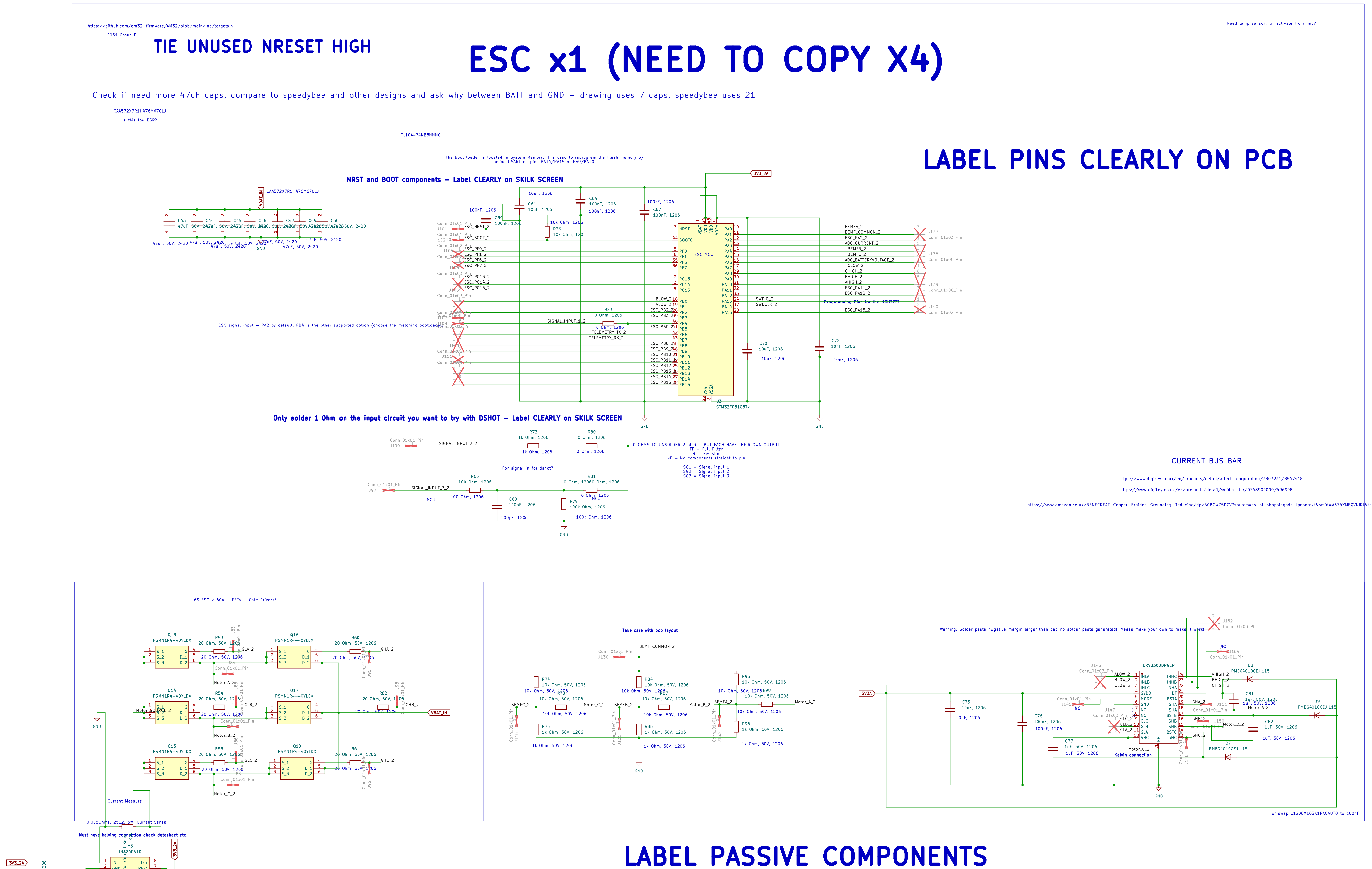
High-power BLDC ESCs often target ≈300–400 µF per 100 A of phase current with low-ESR capacitors placed as close as possible to the MOSFET drain supply.
Seven × 47 µF ≈ 330 µF is in line with a 30–60 A class ESC.

If You Reduce Them

Fewer or lower-value caps → higher voltage ripple → more stress on MOSFETs and driver ICs → possible shoot-through or gate-driver failure.

You might get away with fewer if the current is much lower than the design spec, but it's risky.

```
#ifdef HARDWARE_GROUP_F0_B

#define MCU_F051
#define USE_TIMER_3_CHANNEL_1
#define INPUT_PIN LL_GPIO_PIN_4
#define INPUT_PIN_PORT GPIOB
#define IC_TIMER_CHANNEL LL_TIM_CHANNEL_CH1
#define IC_TIMER_REGISTER TIM3
#define IC_TIMER_POINTER htim3
#define INPUT_DMA_CHANNEL LL_DMA_CHANNEL_4
#define DMA_HANDLE_TYPE_DEF hdma_tim3_ch1
#define IC_DMA_IRQ_NAME DMA1_Channel4_5_IRQn

#define PHASE_A_GPIO_LOW LL_GPIO_PIN_1
#define PHASE_A_GPIO_PORT_LOW GPIOB
#define PHASE_A_GPIO_HIGH LL_GPIO_PIN_10
#define PHASE_A_GPIO_PORT_HIGH GPIOA

#define PHASE_B_GPIO_LOW LL_GPIO_PIN_0
#define PHASE_B_GPIO_PORT_LOW GPIOB
#define PHASE_B_GPIO_HIGH LL_GPIO_PIN_9
#define PHASE_B_GPIO_PORT_HIGH GPIOA

#define PHASE_C_GPIO_LOW LL_GPIO_PIN_7
#define PHASE_C_GPIO_PORT_LOW GPIOA
#define PHASE_C_GPIO_HIGH LL_GPIO_PIN_8
#define PHASE_C_GPIO_PORT_HIGH GPIOA

#define PHASE_A_COMP COMP_PA0
#define PHASE_B_COMP COMP_PA4
#define PHASE_C_COMP COMP_PA5
```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